

Victor Amarante Florentino da Silva

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PROFESSIONAL SUMMARY

Senior AI/ML Engineer with 5 years of experience building production-grade AI systems, grounded in a rigorous foundation of statistics, probability, and applied mathematics — evolved naturally through classical machine learning, deep learning, and into the current frontier of LLM-powered systems and agentic architectures. Specialized in multi-agent systems, RAG and GraphRAG architectures, LLMs including GPT, Gemini, and Claude, and LLMOps, with hands-on experience taking these systems from whiteboard to production at scale across healthcare, fintech, legal, and logistics sectors. Proven ability to move fluidly between applied research and production engineering — with the mathematical depth to understand what happens inside a model and the engineering discipline to build systems that ship and scale. Not an engineer who learned LLM APIs: an ML practitioner who adopted LLMs as the next layer of intelligent systems and never lost sight of the foundations that make AI work for real.

EDUCATION

- Bachelor's Degree in Information Systems** Mar 2023 – Dec 2026
Federal University of Pernambuco (UFPE) - Center of Informatics (CIn) Recife, Brazil
- Bachelor's Degree in Statistics (internal transfer)** Mar 2020 – Feb 2023
Federal University of Pernambuco (UFPE) - Center of Exact and Natural Sciences (CCEN) Recife, Brazil
 - Internal Transfer:** Formal training in probability, applied statistics, linear algebra, and numerical calculus, the mathematical backbone behind machine learning and AI systems. Transferred internally to Information Systems to deepen competencies in Artificial Intelligence, Software Engineering, and System Design.

EXPERIENCE

- Senior AI/ML Engineer** May 2026 – Present
Getronics Amsterdam, Netherlands (Remote)
 - Embedded at one of the largest financial institutions in Brazil as part of a high-seniority squad, owning the full AI model lifecycle, from conception to production monitoring, for a digital assistant in the card services domain.
 - Design and develop autonomous AI agents capable of reasoning and acting, orchestrating LLMs and complex interaction flows using LangChain and LangGraph frameworks.
 - Build and consume REST APIs with MCP and A2A integrations, ensuring seamless connectivity across multiple enterprise systems.
 - Implement cloud-native architectures using Docker, Kubernetes, and cloud platforms, ensuring scalability, performance, and reliability of production AI systems.
 - Drive prompt engineering best practices including context definition, AI safety guardrails, temperature tuning, and token consumption optimization to reduce latency and inference costs.
 - Conduct AI risk assessments and enforce compliance controls across the AI pipeline, including input/output filtering, PII handling, and data privacy requirements in adherence to financial sector regulations.
 - Ensure observability and operational reliability of AI systems through OpenTelemetry.
- Mid-Level AI & Machine Learning Engineer** Sep 2025 – May 2026
Amigo Tech Recife, Brazil
 - Led AI development on the Amigo Flow product, architecting a supervised multi-agent system (orchestrator + specialist agents + dedicated RAG pipelines) applied to the patient journey, delivering a 7.5x increase in automated scheduling and 200%+ growth in confirmation rate, scaling to tens of thousands of monthly appointments.
 - Redesigned the core AI product architecture around a multi-agent pattern, achieving approximately 90% improvement in production response quality and accuracy.
 - Initiated a medical imaging AI pipeline for dental X-ray analysis: performed data annotation, trained YOLO-based object detection models, and built an inference pipeline integrating a VLM to generate structured diagnostic reports in natural language.
 - Built and maintained containerized AI microservices (FastAPI, Docker) with production-grade deployment workflows focused on scalability, reliability, and continuous integration.
 - Engineered prompt lifecycle and governance workflows using version-controlled repositories, applying iterative evaluation, inference optimization, and token consumption analysis to improve quality and reduce operational costs.
 - Implemented conversational monitoring and distributed tracing with LangSmith, enabling behavioral inspection, workflow debugging, and reliability improvements across production agent systems.
- Machine Learning Engineer** Jul 2024 – Sep 2025
C.E.S.A.R – Center for Advanced Studies and Systems of Recife Recife, Brazil
 - Designed and developed a GenAI-powered chatbot and search interface for enterprise knowledge retrieval, combining semantic search with contextual response generation using Django, LangChain/LangGraph, PostgreSQL, and pgvector.
 - Architected and implemented a data integration platform for the transportation/logistics sector: structured an ingestion pipeline on AWS S3 with LLM-based enrichment and assisted knowledge graph construction.
 - Built a production graph database in Neo4j and designed a GraphRAG architecture combining vector search with entity relationship traversal, enabling intelligent queries over previously siloed enterprise data.

- Deployed containerized AI pipelines and production services on AWS (S3, SageMaker, Bedrock) and Google Cloud Platform (Vertex AI), focusing on scalable architectures and enterprise data infrastructure integration.
- Continuously optimized retrieval pipelines, prompting strategies, and conversational workflows to improve response precision, contextual coherence, latency, and inference efficiency.
- Developed APIs and backend services following Hexagonal Architecture and Domain-Driven Design (DDD) principles, adhering to Clean Code and SOLID standards.
- Delivered 20+ hours of in-person data analysis training for SEBRAE analysts (Natal/RN), driving data literacy and tooling adoption across the organization.

• **ML Engineer & Data Scientist**

Nov 2022 – Jun 2024

Queiroz Cavalcanti Advocacia

Recife, Brazil

- Built predictive ML pipelines using Scikit-learn, Pandas, and NumPy for Legal Controlling — including time series forecasting models and classification pipelines — achieving efficiency gains above 90% across 50+ automation and operational intelligence projects in 2023.
- Designed and maintained analytical pipelines, statistical models, and interactive dashboards (Power BI) for strategic KPI monitoring and decision-making support, significantly reducing team operational overhead.
- Conceived and led an internal training program in automation and data science with 80%+ adoption across 40+ professionals on the controlling team.
- Prototyped early generative AI solutions — automated meeting summarization and intelligent email prioritization — during the initial Generative AI adoption wave, establishing the firm's AI foundation.

• **Data Scientist & Statistical Consultant**

Jul 2021 – Jul 2023

Fisher Consultoria Estatística

Recife, Brazil

- Conducted end-to-end statistical consulting projects, from requirements gathering and problem modeling with clients to execution, analysis, and formal report delivery.
- Applied statistical modeling, sampling methods, and exploratory data analysis to solve quantitative problems across different business domains.
- Built data visualizations and analytical dashboards to communicate findings clearly to non-technical stakeholders.
- Managed project scoping, pricing, and client communication across the full consulting lifecycle.

• **Co-Founder & Director of Education**

Dec 2024 – Mar 2026

LIGIA – Academic League of Artificial Intelligence at UFPE

Recife, Brazil

- Structured and coordinated learning tracks in Machine Learning, NLP, and Computer Vision for technical and strategic development of AI practitioners.
- Developed educational content including introductory courses, teaching materials, and audiovisual production for AI training programs.
- Mentored student leaders, fostering a culture of technical excellence, autonomy, and continuous learning.

• **Co-Founder & ML Engineer**

Nov 2023 – Jul 2024

Redduo.AI

Recife, Brazil

- Co-founded and built an AI-powered educational platform for automatic, personalized essay correction for Brazil's ENEM exam, using LLMs and semantic analysis to generate structured student feedback at scale.
- Designed the full data pipeline and backend architecture with cloud service integration and language model APIs, ensuring evaluation consistency and processing scalability.
- Applied NLP, generative AI, and automated text analysis validated with linguistic and semantic metrics using supervised ML techniques.
- **1st place at PoloTec Pre-Incubation (UFPE) and 1st place at PoloTec DemoDay 2024.**

• **Artificial Intelligence Researcher**

Jan 2024 – Mar 2026

GERAIA – Research and Applications Group in Generative AI

Recife, Brazil

- Conducted applied research on LLMs, multi-agent architectures, and RAG systems, with focus on autonomous simulations and distributed intelligent systems.
- Developed AI agents with contextual memory, autonomous reasoning, and generative capabilities for dynamic real-world environments.
- Optimized RAG pipelines and LLM integrations for semantic retrieval, information synthesis, and language-based process automation.

• **Scientific Researcher**

Jul 2023 – Jan 2024

FACEPE – Foundation for Science and Technology Support of Pernambuco

Recife, Brazil

- Conducted research at INES (National Institute for Software Engineering) on accessible mobile applications for individuals with Autism Spectrum Disorder (ASD).
- Applied Human-Computer Interaction (HCI), User-Centered Design, and data analysis methodologies for digital usability and accessibility evaluation.

SKILLS

- **Languages & Software Engineering** Python; TypeScript; FastAPI, Django; Clean Code, SOLID; Hexagonal Architecture, Domain-Driven Design (DDD); unit testing; API documentation; scalable and maintainable code; Git, GitFlow; REST API design
- **AI Agents & Agentic Systems** Multi-agent architectures; supervised agent workflows; orchestrator-specialist patterns; ReAct, self-reflection, hierarchical delegation; CrewAI; LangGraph; Agno; Google ADK (familiar); MCP servers; tool integration; event-driven agent workflows; contextual memory; distributed reasoning
- **LLM Systems & Retrieval Engineering** LLMs (GPT, Gemini, Claude, AWS Bedrock); fine-tuning; RAG and GraphRAG architectures; LangChain; LangGraph; LlamaIndex; Agno; retrieval strategy optimization; chunking and reranking; semantic and hybrid search; conversational memory; contextual orchestration; prompt engineering; language model evaluation
- **LLMOps, Observability & Evaluation** LangSmith; distributed tracing; conversational observability; prompt lifecycle management; inference optimization; token consumption analysis (tokens/sec, cost-per-request); evaluation pipelines; experiment tracking; model monitoring; production AI deployment
- **Cloud, Infrastructure & Deployment** Google Cloud Platform (Vertex AI); AWS (S3, SageMaker, Bedrock); Docker; Docker Compose; containerized AI microservices; API deployment workflows; foundational Kubernetes knowledge
- **Deep Learning & Generative AI** TensorFlow; PyTorch; NLP (spaCy, NLTK); tokenization; embeddings; vector similarity; LLMs; fine-tuning
- **Vector Databases & Search** FAISS; ChromaDB; PGVector; Neo4j (GraphRAG); vector indexing; semantic and hybrid search
- **Machine Learning & Data Science** Predictive modeling; EDA; feature engineering; model selection and validation; hyperparameter tuning; Scikit-learn; MLflow
- **Data Engineering & Analytics** SQL; PostgreSQL; MySQL; SQLAlchemy; Pandas; Amazon Redshift; analytical modeling; Power BI; Tableau
- **Mathematics & Statistics** Probability; Applied Statistics; Linear Algebra; Numerical Calculus; SciPy; Statsmodels

PROJECTS

- **Conectese: Real-Time Public Transit Monitoring Chatbot** Dec 2025
Technologies: Python (FastAPI), Evolution API, LangChain, Groq, React (TypeScript), TailwindCSS 
 - Built a WhatsApp-integrated AI chatbot for real-time monitoring of the Recife bus fleet, improving urban mobility predictability for citizens.
 - Designed a scalable layered architecture (Services, Repositories, Routers) with FastAPI, integrating Evolution API for conversational message flow management.
 - Implemented route and bus stop query systems processing geographic data to deliver accurate arrival estimates and vehicle location in real time.
- **InsightTalk.AI: AI Agent for Data Analysis via Voice Commands** Jan 2025
Technologies: LangChain, OpenAI API, Whisper, TTS, Python, FastAPI 
 - Designed and shipped a voice-interactive conversational AI agent for natural language data exploration, integrating LLMs, LangChain, Whisper (ASR), and TTS.
 - Built a modular architecture with dynamic routes for real-time analytical queries, eliminating technical barriers for non-specialized users.
- **KnowBot: Intelligent RAG Chatbot for Academic Support** Aug 2024
Technologies: LangChain, OpenAI, RAG, Python 
 - Implemented a production-oriented RAG pipeline with PDF semantic ingestion, vector search, and scalable API services for document-based AI dialogue.
 - Designed with architectural scalability and reusability for integration into educational platforms.
- **TalkScribe: AI-Powered Meeting Transcription and Insight Generation** Aug 2024
Technologies: OpenAI Whisper, Streamlit, PyDub, Python 
 - Built a high-accuracy meeting transcription and insight extraction application using OpenAI Whisper, with automatic summarization mechanisms to support post-meeting decision-making.
 - Delivered significant operational efficiency gains by automating manual documentation processes.

LICENSES & CERTIFICATIONS

- **Building Agents with Agno (Asimov Academy)** Aug 2025
- **AI Applications: Multi-Agent Systems with CrewAI (Asimov Academy)** Jun 2025
- **Knowledge Graphs for RAG (DeepLearning.AI)** Mar 2025
- **Developing Chatbots (Rocketseat)** Feb 2025
- **Functions, Tools and Agents with LangChain (DeepLearning.AI)** Sep 2024
- **LangChain for LLM Application Development (DeepLearning.AI)** Sep 2024
- **AI Applications with LangChain (Asimov Academy)** Aug 2024
- **LangChain Chat with Your Data (DeepLearning.AI)** Aug 2024
- **ChatGPT Prompt Engineering for Developers (DeepLearning.AI)** Jul 2024
- **Advanced Microsoft Excel (Udemy)** Jan 2022
- **Advanced Power BI (Udemy)** Jan 2022
- **Statistics with R Language (Udemy)** Sep 2021
- **III Workshop in Applied Statistics (UNIR)** Jul 2020